

Co-assemble: A collaborative AR cross-devices teaching system for assemble practice courses

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ABSTRACT

Assembly trainings in engineering drawing courses mainly rely on physical models, revealing many limitations. We use augmented reality and Azure Spatial Anchors to design "Co-assemble," a multi-user cross-device collaborative system for the assembly practice on mobile devices. There are three modes in the system: single, collaboration, and class. It will help students understanding the models' structure and assembly activities and help teachers easily teach or monitor classes. We also presented and discussed the results of the preliminary user study evaluating the system.

Keywords: Collaborative augmented reality, assemble training, teaching system, multiple devices.

Index Terms: [Augmented Reality]: Spatial Augmented Reality

1 INTRODUCTION

Engineering drawing includes textbook teaching and assembly training. Students will take turns using physical models in assembly training, which limits students' practice time and types. Due to the adjustment of teaching plans in universities, teachers can't guide students' assembly practice in detail, which will not cultivate students' surveying and practical skills. Therefore, we designed a multi-user cross-device collaborative AR system for assembly practice in engineering drawing courses. We first reviewed related work, then described our system architecture and functions. Finally, we do the informal user study and discuss the results.

2 RELATED WORK

Earlier AR assembly training system mainly relies on headsets and computers. Mobile devices became a hot choice for AR systems later. Mika et al.[1] first presented a phone-based AR assembly system. The client-server architecture causes the inability to handle complex models, data transmission delay, and no collaboration. Many studies focus on improving efficiency and accuracy. One solution is to add remote location recording and monitoring[2]. However, unlike worker-oriented training, courses allow or even encourage mistakes. Each step having detailed instructions doesn't help promote students' mastery and creativity.

The potential in education has also been followed. Gojko et al. [3] presented possibilities for integrating AR into engineering drawing training and believed it is the general trend[3]. Xiaofei Wang et al. [4] proposed an AR app, with which users scan the QR code to obtain assembly videos. But it only allows viewing or sample operations and no collaboration. In Villanueva's Meta-AR-App, an AR-based teaching and learning tool [5], the need for

collaboration has been resolved. However, students need to upload information to the cloud, which is delayed.

Overall, with the previous AR assembly systems designed for production, multiple users cannot collaborative and get real-time feedback[6]. And they're not suitable for education. Many efforts for assembly practice in schools are only for single users. Others that support collaboration still lack real-time interaction or rely on devices requiring high costs or cognitive load. They aim to promote the students' learning, and teachers' difficulties have not been well resolved. To provide cross-device collaboration for multi-type users and considering teachers' and students' needs, we used Azure Spatial Anchors (ASA) and AR to design "Co-assemble." It is a cross-device collaborative AR system for assembly in engineering drawing courses. ASA is a cross-platform developer service with which users can create mixed-reality experiences by objects that persist their location across devices over time [7].

3 DESIGN AND IMPLEMENTATION

3.1 System Design

Co-assemble includes a model library, a collaborative module, a user management system, and an interactive module. The model library contains 3D models and information. These are generated by engineering software, so it supports timely updates and mass generation. Students have many practice options, and teachers can use free content and familiar software instead of learning new technology. It is environmentally friendly.

Before class, the teacher selects an object to establish spatial anchors to create a teaching session. Teachers and students can use mobile phones, tablets, and other devices with cameras to link the space anchors together. After registration, users access the authentication system to obtain the matching backend permission. Teachers are administrators, and students are ordinary users. The web service retrieves the virtual models associated with the session from a database (DB) and blob storage. And the virtual models will be presented in proportion to the real size, convenient to survey accurately. Students practice assembly, disassembly, surveying, and complete learning tasks independently or collaboratively. They can also explore freely. All instructions will be archived in the blob storage in file format. The related records in DB will also be updated. At this time, blob storage replays the student's instructions to the teacher client. After class, teachers can use the DB data to summarize error-prone knowledge and improve teaching content.

The system contains three modes of single, collaboration and class, and diversified collaborations in each mode.

Single mode: Users operate on their own devices. It is independent and will not be synchronized to other users in real-time. The single mode is convenient for students to practice according to their thinking and learning rhythm without interference.

Collaboration mode: Students are divided into groups by the teacher. They can operate the model together on their own device or others' devices to complete cooperation tasks. The operations of each group are not synchronized to others in real-time.

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Class mode: Teachers link specific permissions to designated users or groups to operate models and decide whether the changes are synchronized to others in real-time. It will prevent confusion caused by the multi-user simultaneous assembly. The mode is suitable to demonstrate. Teachers can grant students permissions to answer questions, and others watch on their devices. It's similar to answer questions by name, but we make it intuitively and visually.

In single mode, the instructions only change models at the logic level, and the rendering of the models is automatically completed. In collaboration mode, through a backend application, the terminal will broadcast instructions to others in the same "group chat." All instructions received will drive the local model state's change for the viewers, achieving the effect of low data and low latency synchronization. The structures in two modes are both "instruction-logic-render." Too many interface elements will cause confusion, and oral communication is more efficient than online chat, so students communicate in oral language when they collaborate. Also, students' learning skills are different, so we provide different content. In simple mode, the system only provides the needed parts, and in hard mode, the system adds extra parts as interference.

3.2 Implementation

We developed the system in Unity Game Engine version 2019.4. and ran it on iPhone X and Mi9. First, we used SolidWorks to generate assembly models for the model library. With ASA, we generated spatial anchors in the platform. We pulled the students' list, saved it in a DB, and wrote an identity verification system to assign roles and permissions. To support multi-user collaboration, we compiled users' operations into JSON format instructions and wrote a backend application similar to WeChat group chat. Figure 1 shows the student users' interface with seven parts. There will be a novice tutorial for the first use. Users touch the screen according to the specified method for simple operations and touch on-screen buttons to achieve special operations. It has no additional text or audio assembly instructions. Because teachers will demonstrate the assembly before students operating and students can record the process with the "Record" button. If the students click the "Finish" or "Question" button, it means they finish the tasks or have questions, and teachers will receive feedback.

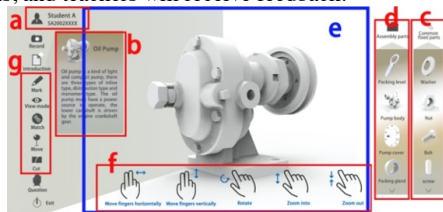


Figure 1: The interface of student users. a. User information; b. Model image and introduction; c. Conventional accessories such as nuts, bolts that can be reused; d. Specific accessories that determined by assembly models; e. Operation area; f. Simple operation: move, zooming in and out, rotate; g. Special operation: match, switch view, section, move along one axis.

4 USERS FEEDBACK

To evaluate the system, we recruited two male teachers and 12 college students (average age=21). Teachers are both experienced in this course. All students' majors require studying engineering drawing courses, including mechanical engineering, automation, and industrial design. We divided eight students who have taken this course to study 1, four students who have not to study 2. The ratio of males to females was the same. Before the studies, the technician instructed how to use the system, and they used their mobile devices to complete the class. We conducted studies in a multimedia classroom. The teacher taught the oil pump and created

spatial anchors on the projection screen. He reviewed the textbook, demonstrated the assembly, and required individual and group tasks. Then, it was free time to practice. Two students did hard learning. Study 2 is the same as study 1, but the teacher taught the assembly's knowledge in more detail. Finally, we interviewed the subjects about the ease of use and the impact of learning assembly and surveying. All users in study 1 affirmed our system's superiority in class: easy to use, help observe and assemble, reduce measurement errors, ample operating time. All inexperienced subjects thought the system was easy and fun. They also proposed that personalized learning content wasn't various, touch screen function was too susceptible, and the delicate parts cannot be operated well. Two teachers agreed that the system could manage the classroom situation well. They suggested that if multiple students "Question" together, it may be challenging to handle.

5 CONCLUSION AND FUTURE WORK

The study presented the possibility of multi-user cross-device collaboration based on AR to assist assembly course. The system takes care of teachers and students and uses mobile devices to make it very convenient. Although the system is suitable for most mobile devices, it has been tested to work best on tablets. Compared with the traditional assembly, we miss the perception of weight and volume but avoid the surveying errors caused by worn models and increase models type. Comparing with 3D software such as CAD, the operations is more intuitive and contains context awareness, which helps obtain critical knowledge in the physical environment. To other AR assembly systems, we provide low-latency cross-device collaboration. The interaction between teachers and students will help teach specifically and learn efficiently. It can be said that our system is a solution between the above three. The users' evaluation is positive overall, and we believe that it is also easy to use for people without related course experience. Next, we will explore how to avoid problems caused by touching the screen and use multi-modal information to support remote communication. Formal user research and pilot projects also are necessary. We also consider designing a collaborative mode operating in the LAN to reduce communication pressure with the external network. WEB service will be integrated into a particular client, sending and receiving commands to devices in the LAN in UDP packets.

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